

Digital Payment Wallet System in Java

Aim / Objective

To design and implement a Digital Payment Wallet System in Java using an interface, simulating real-world digital payment operations such as wallet creation, money addition, UPI payments, balance checking, and transaction validation.

Description

The Digital Payment Wallet System creates a wallet for a user and performs basic digital payment operations. The program uses an interface for payment services, encapsulation for wallet data, and custom exceptions to validate UPI IDs, transaction amounts, and wallet balance before processing a payment.

System Requirements

Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm

1. Start the program.
2. Create the PaymentService interface with pay() and checkBalance().
3. Create the Wallet class with private user details and balance.
4. Initialize wallet details using the constructor.
5. Define addMoney(), getBalance(), displayWalletDetails(), and deductMoney().
6. Create UPIPayment and implement PaymentService.
7. Validate the receiver UPI ID before processing the payment.
8. If the UPI ID is invalid, throw InvalidUPIException.
9. Validate the payment amount and throw InvalidAmountException for zero or negative amounts.
10. Check the wallet balance and throw InsufficientBalanceException when the balance is insufficient.
11. If all validations are successful, deduct the payment amount and display the success message.
12. Use try-catch-finally to handle transaction exceptions.
13. Display the final wallet balance and stop the program.

UML Diagram

PaymentService + pay() + checkBalance()	UPIPayment - wallet + pay() + checkBalance()	Wallet - userName - mobileNumber - upid - balance + addMoney() + getBalance() + deductMoney()
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Source Code

```
interface PaymentService {
    void pay(String receiverUPI, double amount)
        throws InvalidUPIException, InvalidAmountException,
        InsufficientBalanceException;
    void checkBalance();
}

class Wallet {
    private String userName;
    private String mobileNumber;
    private String upiId;
    private double balance;

    Wallet(String userName, String mobileNumber,
            String upiId, double balance) {
        this.userName = userName;
        this.mobileNumber = mobileNumber;
        this.upiId = upiId;
        this.balance = balance;
    }

    void addMoney(double amount) throws InvalidAmountException {
        if (amount <= 0)
            throw new InvalidAmountException("Invalid amount");
        balance = balance + amount;
        System.out.println("Money added: " + amount);
    }

    double getBalance() {
        return balance;
    }

    void displayWalletDetails() {
        System.out.println("User Name : " + userName);
        System.out.println("Mobile Number : " + mobileNumber);
        System.out.println("UPI ID : " + upiId);
        System.out.println("Balance : " + balance);
    }

    void deductMoney(double amount) {
        balance = balance - amount;
    }
}

class UPIPayment implements PaymentService {
    private Wallet wallet;

    UPIPayment(Wallet wallet) {
        this.wallet = wallet;
    }

    public void pay(String receiverUPI, double amount)
        throws InvalidUPIException, InvalidAmountException,
        InsufficientBalanceException {

        if (!receiverUPI.contains("@") ||
            receiverUPI.startsWith("@") ||
            receiverUPI.endsWith("@"))
            throw new InvalidUPIException("Invalid UPI ID");

        if (amount <= 0)
            throw new InvalidAmountException("Invalid payment amount");

        if (amount > wallet.getBalance())
            throw new InsufficientBalanceException(
                "Insufficient wallet balance");

        wallet.deductMoney(amount);
        System.out.println("Payment successful");
        System.out.println("Paid: " + amount);
        System.out.println("To: " + receiverUPI);
    }

    public void checkBalance() {
        System.out.println("Available Balance: " + wallet.getBalance());
    }
}

class InsufficientBalanceException extends Exception {
```

```

        InsufficientBalanceException(String message) {
            super(message);
        }
    }

    class InvalidUPIException extends Exception {
        InvalidUPIException(String message) {
            super(message);
        }
    }

    class InvalidAmountException extends Exception {
        InvalidAmountException(String message) {
            super(message);
        }
    }

    public class DigitalPaymentDemo {
        public static void main(String[] args) {
            Wallet wallet = new Wallet(
                "Rahul", "9876543210", "rahul@upi", 5000);

            UPIPayment payment = new UPIPayment(wallet);
            wallet.displayWalletDetails();

            System.out.println("\n===== TRANSACTIONS =====");
            try {
                wallet.addMoney(2000);
                payment.pay("ravi@upi", 1500);
                payment.checkBalance();
                payment.pay("wrongupi", 500);
            } catch (InvalidUPIException |
                InvalidAmountException |
                InsufficientBalanceException e) {
                System.out.println("Transaction failed: "
                    + e.getMessage());
            } finally {
                System.out.println("Transaction process completed.");
            }

            System.out.println("\n===== FINAL WALLET =====");
            wallet.displayWalletDetails();
        }
    }
}

```

Compilation and Execution Commands

Save as: DigitalPaymentDemo.java
Compile: javac DigitalPaymentDemo.java
Execute: java DigitalPaymentDemo

Output Screen – Wallet Creation and Successful Payment

```
DigitalPaymentDemo.java

C:\> javac DigitalPaymentDemo.java
C:\> java DigitalPaymentDemo

User Name : Rahul
Mobile Number : 9876543210
UPI ID : rahul@upi
Balance : 5000.0

===== TRANSACTIONS =====
Money added: 2000.0
Payment successful
Paid: 1500.0
To: ravi@upi
```

Output Screen – Invalid Transaction Handling

```
DigitalPaymentDemo.java

C:\> java DigitalPaymentDemo

===== TRANSACTIONS =====
Money added: 2000.0
Payment successful
Paid: 1500.0
To: ravi@upi
Available Balance: 5500.0
Transaction failed: Invalid UPI ID
Transaction process completed.

===== FINAL WALLET =====
Balance : 5500.0
```

Result

Thus, the Digital Payment Wallet System was implemented successfully to demonstrate interfaces, encapsulation, String validation, custom exceptions, exception handling, wallet transactions, and UPI payment processing using Java.